Aadish Verma

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Founding Developer, JV Copilot (2025 - present)

- I work with JV Copilot, a startup building a product which abstracts the management of joint-venture marketing campaigns.
- Within our codebase, which is a React/TypeScript codebase, I work primarily on our backend but also often touch the frontend. My work involves adding requested features and expanding our product surface, as well as various improvements and refactors for our technical architecture, such as error handling, logging, and performance. Our codebase currently has over 100,000 lines of code.

Software colead, Project Pixel Orbital (2025 - present)

- I am the software colead for Project Pixel Orbital, a team of students at Stanford Online High School designing and building a 3U CubeSat targeting launch into low-Earth orbit NET early 2027. We are the first fully student-led CubeSat team to have no adult supervision.
- All of our firmware is written in no_std Rust using the RTIC framework for concurrency on an STM32H753ZI microcontroller. My work includes writing drivers for our various peripherals and creating codebase-wide primitives for logging and serialization. As part of this, we have worked with professors from UC Berkeley, students from Stanford University, technical employees at the Jet Propulsion Laboratory, and former members of the FOSSASAT-1 project.

VEX Robotics (2023 - 2025)

- I was the lead programmer on VEX V5 Robotics Competition team 315P. I worked on creating fast and consistent autonomous routines for our robot. I also designed abstractions around control libraries which used modern C++ features so new programmers could onboard and start writing code quickly. We were division quarterfinalists at the VEX Over Under world championship.
- I led AI programming on the VEX AI Robotics Competition team 3151A. VEX AI differs from other VEX competitions because robots move entirely autonomously, with no driver control, for the entirety of the match. My work on 3151A involved writing a custom computer vision stack and associated tooling in Python and C++ which inferenced a fine-tuned YOLOv5n model using TensorRT on a NVIDIA Jetson Nano. As part of my exploratory work on the team, I also designed a system using an RPLiDAR A1 sensor to perform Monte Carlo Localization on a Jetson Orin Nano. This system involved a Rust node (with statically linked C++) communicating with a Python node over the Zenoh protocol, enabling concurrent and GPU-accelerated localization. We were world quarterfinalists at the VEX High Stakes world championship.

Open-source and community work

- I have contributed to commonly used libraries in the VEX ecosystem, such as vexide and LemLib. I also am one of the primary contributors to Venice, an open-source Micropython runtime for the VEX platform written in embedded Rust.
- I now no longer compete in VEX, but I maintain a blog with thousands of monthly pageviews and write about complex topics in control theory, including Monte Carlo localization, Kalman filters, motion algorithms, and more. Several teams have used my blog posts as a reference when implementing these ideas themselves.

Sophomore, Stanford Online High School

- My coursework includes Linear Algebra, Differential Equations, and Physics C.